



Audit Report 2024

In accordance with the following requirements:

Puro.earth - Biochar Methodology

Groupe Bordet Four Bordet
21290 Leuglay
Operator's No.: PE-71059

Contact details operator

Name and address

Four Bordet Groupe Bordet
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Contact person(s)

Monsieur Cyril Flores

Audit visit details

Date

10.09.2024

Duration

2 h 15 m

Persons present including their function

ALBAREL Charlotte, General Director
FERRIERE Nicolas, Operations Manager Carbon App
GARNIER Bulle, Associated member Carbon App
DESHOUX Maëlle, R&D Project responsible
TUAL Gabriel, Associated member Carbon App
Philipp Seitz, bio.inspecta AG, Auditor

	very good			not satisfactory		
Clarity of documentation	☐	☐	☐	☒	☐	☐
Audit visit preparation:	☐	☐	☐	☒	☐	☐

O.K. ☐ Corrective action required ☐
 Not verified ☐ Not relevant ☐

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				1	Audit Description
☒	☐	☐	☐	1.01	Audited Standard: <i>Puro.earth CO2 Removal Marketplace General Rules 3.1 – Biochar Methodology (Annex A)</i>
☒	☐	☐	☐	1.02	Type of Audit: <i>Output Audit</i>
☒	☐	☐	☐	1.03	Auditing Body: <i>bio.inspecta AG, Ackerstrasse 117, CH-5070 Frick www.bio-inspecta.ch</i>
☒	☐	☐	☐	1.04	Audit order assigned to an impartial auditor, free from any conflicts of interest, capable and qualified to complete this audit according to Puro Standard. <i>Auditor (name/surname): PHILIPP SEITZ</i>
☒	☐	☐	☐	1.05	Audit ID: <i>PE-71059</i>
☒	☐	☐	☐	1.06	Audit Date: <i>10.09.24</i>
☒	☐	☐	☐	1.07	Production Facility Location: <i>Froidvent, 21290 Leuglay</i>
☒	☐	☐	☐	1.08	Production period: <i>01.05.2023 - 30.04.2024</i>
☒	☐	☐	☐	1.09	Audit could be finished within the scheduled time frame
				2	Standing Data Confirmation

O.K
Corrective action required
Not verified
Not relevant

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				2	Standing Data Confirmation
☒	☐	☐	☐	2.01	The standing data has been collected from Puro and checked for consistency against other evidence. (GL Ref.1.2.5.) <i>Trade registry available; location evidenced; removal method eligible; no public support. Evidence of output volume: Consolidated protocol shows production of 3.755 t during 12 month crediting period (01.05.23 - 30.04.24). Two EBC batches overlap with crediting period: (1) 5 month overlap with ba-fr-115-1-2 from 01.05.23 to 21.09.23; (2) 7 month overlap with ba-fr-1151-1-3 (22.09.23 - 30.04.24). Minor quantities sold during the second validation period were produced during the first crediting period (and allocable to ba-fr-115-1-1). Total quantities produced are significantly superior to quantities claimed as biochar. 519 t wet provenly sold as biochar, equivalent to 13.82%.</i>
				3	Evidence Confirmation
☒	☐	☐	☐	3.01	All necessary evidence has been provided to the auditor by the Production facility and has been used to complete the compliance checklist. (GL Ref. 5.) <i>Proof of product quality based on EUROFINS laboratory analyses linked to to the respective batches in which eligible volumes were produced (AR-21-FR-039159-01 for ba-fr-115-1-1; AR-23-FR-008917-01 for ba-fr-115-1-2 and EX-23-FR-003349-01 for ba-fr-115-1-3; Evidence of output volume: Consolidated protocol shows production of 3.755 t during 12 month crediting period (01.05.23 - 30.04.24). Two EBC batches overlap with crediting period: (1) 5 month overlap with ba-fr-115-1-2 from 01.05.23 to 21.09.23; (2) 7 month overlap with ba-fr-1151-1-3 (22.09.23 - 30.04.24). Minor quantities sold during the second validation period were produced during the first crediting period (and allocable to ba-fr-115-1-1). Total quantities produced are significantly superior to quantities claimed as biochar. 519 t wet provenly sold as biochar, equivalent to 13.82%. Proof of no double counting: written declaration of buyers available; Carbon Credit Withdrawal Right issued, thereby transferring the right to generate carbon credits from the carbon content of the biochar (example attached).</i>
				4	Eligibility Checklist

O.K
Corrective action required
Not verified
Not relevant

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				4 Eligibility Checklist
☒	☐	☐	☐	4.01 Biochar is used in applications other than energy. (GL Ref. 1.1.1.) <i>58% of sold biochar is used as soil amendment (mainly in compost). The remainder is mostly used for animal nutrition. Biochar used as material and for other applications is negligent. 80% of bulk sales to major customers have been cross-checked for its end use (Sylva Fertilis; Florentaise; Callegari), attesting to the non-energetic end-use.</i>
☒	☐	☐	☐	4.02 Biochar is produced from sustainable forest or waste biomass raw materials (consult positive list of biomasses). (GL Ref. 1.1.2) <i>23% of the sourced raw material is PEFC certified, while only 13.8% of production is sold as biochar. From a perspective of mass flow, all biochar is PEFC certified. More production has been done with PEFC certificates (23%) than the production of eligible biochar (13.82%). PEFC certificates available for 22 suppliers.</i>
☒	☐	☐	☐	4.04 Pyrolysis reactor input fuel for heating is not a fossil fuel. Unless only used for ignition/pre heating or in a mobile unit and the emissions are fully included in the LCA. The use of waste heat from other industrial processes (eg. Biodigesters, cement production) is permitted. (GL Ref. 1.1.4.) <i>One single ignition w/ gas (3-5 hours) and fuel (5-6 hours) per year following maintenance. The rest of the year the pyrolysis operates auto-thermal w/ synthesis gases and tar (both generated during the pyrolysis).</i>
☒	☐	☐	☐	4.05 Pyrolysis gases are combusted or recovered. Bio-oil and pyrolysis gases can be stored for later use as renewable energy or materials. (GL Ref. 1.1.5.) <i>Gases are trapped and tar separated. The clean gas is partly used to produce electricity for own consumption, partly to fuel the boiler, the same as condensed and separated tar which is sprayed into it. The generated heat is directed towards the dryers.</i>
☒	☐	☐	☐	4.06 The molar H/Corg ratio is less than 0.7. <i>The molar H/Corg ratio varies between 0.22 to 0.25 according to the respective analytical reports.</i>

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Page 6 of 16

O.K
Corrective action required
Not verified
Not relevant

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				5	LCA Checklist
☐	☐	☐	☒	5.01	LCA complete and shows: carbon footprint of the biomass production and supply , emissions from the biochar production process , carbon footprint of the biochar end use - cradle to grave. (GL Ref. 1.1.3) <i>Emissions A1 – A4, B1 & Infrastructure included. Application is not included in</i> <i>Euse for 2 reasons: (1) compost is considered by all biochar methodologies as a stable sink matrix; (2) for compost production (58% of sales) and animal feed (48% of sales) no additional energy input is required as compared to the baseline; Requirement 1.1.6: One</i> <i>single ignition w/ gas (3-5 hours) and fuel (5-6 hours) per year following maintenance. The rest of the year the pyrolysis operates w/ synthesis gases and tar (both generated during the pyrolysis); 1.1.7: Gases are trapped and tar separated. The clean gas is partly used to produce electricity for own consumption, partly to fuel the boiler, the same as condensed and separated tar which is sprayed into it; 1.1.8: The</i> <i>generated heat is directed towards the dryers; 1.1.9 Carbon content provided as part of the respective EUROFINS analytical reports; 1.10: The molar H/Corg ratio varies between 0.22 to 0.25 according to the respective analytical reports; 1.1.11 The molar O/Corg ratio significantly below 0.4 across the respective analyses.</i>
☒	☐	☐	☐	5.02	The CO2 Removal Supplier provides a life cycle assessment (LCA) for biochar activity including disaggregated information on the emissions arising at different stages. The system boundary is set cradle-to-grave and includes emissions from production and supply of the biomass, from biomass conversion to biochar, and from biochar distribution and use. (GL Ref. 3.1) <i>LCA provided by CarbonApp. The system boundary of the LCA was set to cradle-to-consumer, consisting of biochar production, including raw material supply/transport and distribution within a 100-year time boundary. Infrastructure is included. Application has not been factored into</i> <i>Euse for 2 reasons: (1) compost is considered by all biochar methodologies as a stable sink matrix; (2) for compost production (58% of sales) and animal feed (48% of sales) no additional energy input is required as compared to the baseline.</i>

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				5	LCA Checklist
☐	☐	☐	☒	5.04	The default baseline emission scenario for the project activity feedstock is zero, which is a conservative assumption since it is not taking into account methane emissions derived from decay of manure or combustion of waste biomass. If a non-zero baseline presented, needs to be accepted by Puro.earth <i>The default baseline emission scenario is assumed to be zero.</i>
☒	☐	☐	☐	5.03	Life cycle assessment (LCA) follows ISO standard, WRI GHG protocol or similar method. (GL Ref. 3.2) <i>The LCIA was conducted using CML -IA baseline (version 3.05) for the characterization factors of GWP. The CML -IA baseline method uses the characterization factors as developed by the Intergovernmental Panel on Climate Change (IPCC), expressed as GWP for time horizon 100 years (GWP100). According to the SFS-EN ISO 14040 standard series, the LCIA includes both mandatory and voluntary parts. The LCI is further based on a CO2 footprint calculation conducted for the period 09/2019 to 08/2020 (12 months).</i>
				6	Production Facility Checklist (Desktop and Verbal Confirmation).

O.K
Corrective action required
Not verified
Not relevant

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				6	Production Facility Checklist (Desktop and Verbal Confirmation).
☐	☒	☐	☐	6.01	Evidence of Production Facility eligibility under the general rules of Puro Standard. (GL Ref. 1.2.1) <i>The produced biochar is not used for energy purposes. Biochar produced during the period has been marketed and sold as input for composting and animal feed. (§1.1.1). The feedstock consists exclusively of wood chips. (§1.1.4.) Feedstock supply for the wood is well documented and volumes certified according to PEFC proportionally higher than sold volumes (28% to 13.82%). (§1.1.2). The facility is EBC-certified. (§1.1.7) Compliance with comparable conditions is outlined in this report, and all process emissions have been calculated in the LCA using the cradle-to-grave approach except for harvesting and end-use application (for justification see above). The following emissions have been accounted for: Transport of wood to the production site; chipping; the manufacturing of the biochar on site and transportation of the biochar; harvesting is accounted for in the Carbon Balance Sheet of the sawmills supplying the Bordet Group. Four Bordet was authorized to exclude harvesting data in the carbon accounting of the activity after submission of sworn certificates of accounting by sawmills; however, not all saw mills did provide a sworn certificates. (§1.1.3). Pyrolysis gases are captured and combusted within the reactor to maintain the carbonisation process. (§1.1.5). The remaining gases are combusted and exhausted. The exhaust emissions have been analysed and quantified. (§1.2.4). The produced biochar has a high, stable carbon content. The lab analyses demonstrated an organic carbon content between 88.7% to 93.7% in the dry state. (§5.3.3). The produced biochar has a H/Corg molar ratio of 0.22 to 0.25, well below the 0.7 threshold. (§1.1.8). No quenching is done. Although intended for agricultural purposes, biochar leaves the compound between 4.1 and 7.6% instead of the recommended minimum of 20%, let alone the reported 30% which was taken as granted for classification and allocation of the EBC quality status. (§1.1.8). The potential for methane emissions from stockpiling of wet feedstock onsite and at the supplier has been considered in accordance with the criteria of the EBC. There are no emissions from feedstock storage. The potential for fugitive release of syngas from pyrolysis is considered negligible.</i>

O.K.
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				6	Production Facility Checklist (Desktop and Verbal Confirmation).
☒	☐	☐	☐	6.02	The Production Facility demonstrate Environmental and Social Safeguards. (GL Ref. 1.2.2.) <i>The LCIA process characterises and assesses the effects of environmental releases identified in the LCI into impact categories such as global warming, acidification, carcinogenics, respiratory effects, eutrophication, ozone depletion, ecotoxicity, and smog. The global warming impact category is the only one that must be considered.</i>
☒	☐	☐	☐	6.03	CO2 Removal Supplier shall be able to demonstrate additionality, meaning that the project must convincingly demonstrate that the CO2 removals are a result of carbon finance. Even with substantial non-carbon finance support, projects can be additional if investment is required, risk is present, and/or human capital must be developed. To demonstrate additionality, CO2 removal Supplier must provide full project financials and counterfactual analysis based on Baselines that shall be project-specific, conservative and periodically updated. Suppliers must also show that the project is not required by existing laws, regulations, or other binding obligations. (GL Ref. 1.2.3) <i>Additionality requirements fulfilled. Additionally statement attached to the validation order.</i>
☒	☐	☐	☐	6.04	The Production Facility's documentation system is accurate and reliable (GL Ref. 1.2.4) <i>Evacuations of biochar from the kiln is computed. One evacuation corresponds to 96 kg. The total number of evacuations during a batch period corresponds to the total annual amount, minus 52 evacuations corresponding to one kiln which is eliminated after the annual maintenance stop (app. 5 t). Each biomass delivery is weighed and recorded. Electricity is metered (invoice attached). Invoices for propane and fuel consumption attached.</i>

O.K
Corrective action required
Not verified
Not relevant

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				6	Production Facility Checklist (Desktop and Verbal Confirmation).
☒	☐	☐	☐	6.05	The quantity of the biochar produced and sold is quantified and documented in a reliable manner (GL Ref. 1.2.4) <i>Evacuations of biochar from the kiln is computed. One evacuation corresponds to 96 kg. The total number of evacuations during a batch period corresponds to the total annual amount, minus 52 evacuations corresponding to one kiln which is eliminated after the annual maintenance stop (app. 5 t). Sales records complete and transparent (attached).</i>
☒	☐	☐	☐	6.06	Relevant meters are in place and they are calibrated (GL Ref. 1.2.4) <i>Calibration certificates of weighing bridge submitted (attached).</i>
☒	☐	☐	☐	6.07	The emissions from the cultivating, harvesting and transporting of the biomass are estimated and calculated in a reliable manner (GL Ref 1.2.4) <i>Cultivation and harvesting is accounted in the Carbon Balance Sheet of the sawmills supplying the Bordet Group (see sworn certificate of counting). Four Bordet was therefore authorized to exclude harvesting data in the carbon accounting of the activity. Transport considered. Total biomass considered for wooden edgings and short wood chutes. For each delivery, truck category, tonnage (net weight wet) and transport distances are indicated. Through multiplication of tonnage and distance tonkm are being generated which is the required unit to apply the truck category-specific emission factors. Two truck categories and respective emission factors can be distinguished (see Bilan Carbone). Calculations rely on supplier invoices.</i>
☒	☐	☐	☐	6.08	The energy use of the Production Facility can be quantified and the emissions from the process calculated (GL Ref. 1.2.4) <i>Electricity is metered (invoice attached). Invoices for propane and fuel consumption attached.</i>
☒	☐	☐	☐	6.09	The auditor goes through the Quantification of CO2 Removal requirements with the CO2 Removal Supplier, so that the Supplier is able to calculate the CO2 Removal independently in its Output Report <i>Not with the CO2 Removal Supplier but with the LCA provider</i>
				7	Calculation Checklist

O.K
Corrective action required
Not verified
Not relevant

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				7	Calculation Checklist
☒	☐	☐	☐	7.01	Qbiochar = Quantity of biochar produced and sold to end user. (dry char) (GL Ref. 4.2.) <i>3.755 t dry produced and 519 t dry sold as biochar.</i>
☒	☐	☐	☐	7.02	FpTHTs = $c + m \times H/C_{org}$ (GL Ref. 4.2.) <i>Provided in the Gross embodied CO2 calculator at given soil temperature and selected time horizon; varies between 0.90 - 0.91 depending on the batch and the corresponding analyses associated.</i>
☒	☐	☐	☐	7.03	C Biochar = carbon content of biochar (GL Ref. 4.2.) <i>The organic carbon content of the biochar was measured between 88.7 and 93.8% in the respective analyses corresponding to the different batch periods, in the dry state. Therefore, the mass of captured CO2 contained in each dry mt of biochar varie between 1,91 and 3,13 tonnes CO2/tonne of biochar.</i>
☒	☐	☐	☐	7.04	Estored = biochar carbon storage = Qbiochar x Cbiocharorg x FpTHTs x 44/12 (GL Ref. 4.2.) <i>The laboratory test shows that the biochar has a H/Corg ratio of 0.22-0.24, indicating a good level of stability. Using the combination of H/Corg ratio and an average soil temperature of 14.4°C, in mainland France, it has been determined that between 90-91% of biochar is durable at 100 years. A 9-10% buffer has therefore been factored into the carbon removal calculation per the Puro.Earth methodology for biochar.</i>
☒	☐	☐	☐	7.05	Ebiomass = LCA emissions of production and supply of biomass (GL Ref. 4.3.) <i>The Ebiomass calculation shows that 33.6 tonnes of CO2e emissions are caused by the wood used to produce the biochar during the period. The allocated emissions amount to 67 kg CO2e/tonne of dry biochar.</i>

O.K
Corrective action required
Not verified
Not relevant

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				7	Calculation Checklist
☐	☒	☐	☐	7.06	Eproduction = LCA emissions from biochar manufacturing (GL Ref. 4.4) <i>Emissions from the production process of the biochar are limited to electricity (for the whole site), diesel, GNR and fuel, as well as process emissions. Moreover, allocated emissions arise from the infrastructure. In total, the allocated emissions of greenhouse gases related to the manufacturing of biochar amount to 97 kg CO2e per tonne of dry biochar.</i>
☒	☐	☐	☐	7.07	Euse = LCA emissions of the use of biochar, including distribution up to the point of final use (GL Ref 4.5) <i>The bagged biochar is transported to the customer by truck. The volume-weighted average distance travelled for biochar to customers has been calculated, greenhouse gas emissions from biochar transport is therefore based on tonkm transported. The transport to end user has been modelled using the Ecoinvent dataset "transport, freight, lorry >32 metric ton, EURO6 - RER". Total biochar transport emissions, including the biochar and packaging mass, are 15 kg CO2e per tonne of dry biochar. Application is not included in</i> <i>Euse for 2 reasons: (1) compost is considered by all biochar methodologies as a stable sink matrix; (2) for compost production (58% of sales) and animal feed (48% of sales) no additional energy input is required as compared to the baseline.</i>
☒	☐	☐	☐	7.08	CORCs = Estored - Ebiomass - Eproduction - Euse <i>For 01.05.23 – 30.04.24, the net carbon removal is 2,824 tonnes CO2e/t of dry biochar.</i> <i>CORCs: 1.403,373</i>
☒	☐	☐	☐	7.09	Quantity of CORCs (in evidence). <i>Given the mass of 496 dry tonnes sold during the assessed period, the resulting total CO2e removed was 1403,5 t CO2.</i>

O.K
Corrective action required
Not verified
Not relevant

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				7	Calculation Checklist
☒	☐	☐	☐	7.10	Confirm consistency. <i>All inconsistencies have been solved with the responsible LCA provider during the validation exercise.</i>
				9	Overall conclusion
☒	☐	☐	☐	9.01	Overall conclusion: <i>The validator confirms that the LCA calculation for the validation period (01.05.23 – 30.04.24) provides a credible and faithful account of output volumes and emissions, and thus of declared carbon dioxide removals eligible for CORCs as stated in the Output statement.</i>

Auditor's evaluation and recommendation

Non-compliance	Corrective action	Deadline
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4.08 No quenching is done. Although intended for agricultural purposes, biochar leaves the compound between 4.1 and 7.6% instead of the recommended minimum of 20%, let alone the reported 30% which was taken as granted for classification and allocation of the EBC quality status.	Please provide justification why biochar intended for agricultural purposes requires no quenching and explain which safety measures are implemented to safeguard transport (this aspect will be dealt with in the framework of EBC certification later this month).	
6.01 Four Bordet was authorized to exclude harvesting data in the carbon accounting of the activity after submission of sworn certificates of accounting by sawmills; however, not all saw mills did provide a sworn certificate. Sworn certificates are missing for the following saw mills: MAILLARD, HAUTRIVE, PROVENCHERE, SOUPEAUX, SANCHEZ, DELTA BOIS, BCFC, SEFIC, BOURGOGNE, JEUNEUX, CARNIO.	Please submit a written confirmation that missing sworn certificates will be submitted in the short term (submission of sworn certificates will be checked during next validation period, and harvesting and transport emissions eventually accounted for proportionally and the CORC factor rectified).	19.09.2024

The Right to be Heard

The undersigned has reviewed the outcome of the audit documented in this report and confirms the completeness and accuracy of the information provided in the audit and the content of this report.

He/ she has taken note of the non-conformities, measures, deadlines and sanctions described in this report.

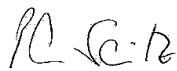
The undersigned has the option of submitting a counter-notification in writing to bio.inspecta AG within three working days of receipt of this report. If no reply is received within this period, the contents of this report shall be deemed to be acknowledged.

Frick, 21.09.2024

Leuglay,

bio.inspecta AG / q.inspecta GmbH
International Department

Groupe Bordet Four Bordet



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Philipp Seitz

name, first name.....

Auditor

function.....